MIL-A-14838(MU)

19 September 1968
SUPERSEDING
FA-PD-MI-2395, Rev. 0
29 July 1960

MILITARY SPECIFICATION

ARM-SAFE DEVICE M35, LESS SETTING MECHANISM ASSEMBLY

1. SCOPE

1.1 This specification covers one type of electro-mechanical Arm Safe Device, Less Setting Mechanism Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-S-45743 - Soldering Electrical Connections, for Guided and Ballistic Missiles, with Electrically-Heated Soldering Irons, and Resistance Soldering Apparatus Procedures for

STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-331 - Fuze and Fuze Components Environmental and Performance Tests for
MIL-STD-1167 - Ammunition Data Card
MIL-STD-1169 - Packaging, Packing and marking for Shipment of Inert Ammunition Components

FSC 1390

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DRAWINGS

Munitions Command

E8596720 - Arm Safe Device M-35, Less Setting Mechanism Assembly

IEL-8596720 - Inspection Equipment List for Arm Safe Device, M-35, Less Setting Mechanism Assembly

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASA-B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y.)

3. REQUIREMENTS

3.1 General. - Materials, parts, and assemblies shall comply with the requirements specified on the applicable drawings, referenced standards and specifications, except as provided in 6.1.

3.1.1 Plastic Materials.

3.1.1.1 Thermal shock. - The plastic rotors and covers shall withstand thermal shock without fracturing, cracking or deforming when tested in accordance with MIL-STD-331, test No. 113.

3.2 Resistance (Switch Assy)

3.2.1 Insulation resistance (with all switches closed). - The insulation resistance between the following terminals: 1 & 2, 1 & 3, 1 & 4, 2 & 3, 2 & 4, and 3 & 4 shall be 100 megohms minimum, when measured at 500 $\pm$ 50 volts, dc, with current limited to 5 milliamperes max.

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3.2.2 Circuit resistance. -The circuit resistance of the three switches and all combinations of two of the three switches through each closed circuit shall not exceed 0.3 ohms maximum under the following conditions: open circuit voltage 5.5 + 0.5 volts, dc; current 100 ± 5 milliamperes (switches 1, 2, & 3 closed. Switch Combination: Switches 1 and 2 closed, Switches 1 and 3 closed and Switches 2 and 3 closed. Circuits: 1-8, 2-7, 3-5, and 4-6).

3.3 Movement Assembly

3.3.1 Static regulation. -The movement assembly shall be regulated as specified on Drawing F8596718 and shall have an accuracy within ± 0.02 seconds at any time point during each of two consecutive runs of 50 seconds.

3.3.2 Nonarming (with setback safety leaf released). -When subjected to an acceleration of 600g, minimum, the escapement lever lock shall not release.

3.3.3 Arming (with setback safety leaf released). -When subjected to an acceleration of 860g, maximum, the escapement lever lock shall release.

3.3.4 Function (with setback safety leaf released). -When subjected to an acceleration of 1150 to 1300g for each of two consecutive runs, the setback mechanism shall arm, movement shall run and firing arm shall release.

3.4 Arm Safe Device, Less Setting Mechanism

3.4.1 Function. -The device shall function within the setting range of 1 to 50 seconds and within the temperature range of minus 40 degrees to plus 125 degrees Fahrenheit (F) and shall demonstrate the required electrical properties.

3.4.1.1 Safety. -The device shall be incapable of functioning at any time setting from 0 to 0.5 seconds. The device shall not function prematurely (See 6.4).

3.4.1.2 Timing accuracy. - Within the specified temperature range, the device shall have an accuracy within plus or minus 0.25 seconds at any time setting 1 to 4 seconds, and plus 0.60 seconds and minus 0.25 seconds at any time setting 5 to 50 seconds.

3.4.2 Low acceleration (with setback safety leaves released). -When subjected to an acceleration of 1150 to 1300g the setback mechanisms shall arm, movements shall run, firing arms shall release, the switches shall complete circuit continuity and the device shall comply with the requirements as specified in 3.4.1.2 and 3.4.7.

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3.4.3 High acceleration (with setback safety leaves released).-When subjected to a high acceleration of 2500 to 2750g, the setback mechanisms shall arm, movements shall run, firing arms shall release, the switches shall complete circuit continuity and the device shall comply with the requirements as specified in 3.4.1.2 and 3.4.7.

3.4.4 Thermal shock.-The device shall withstand the thermal shock when tested in accordance with MIL-STD-331, Test No. 113, and shall comply with the requirements of 3.4.2.

3.4.5 Vibration.-The device, in a fixture, shall show no evidence of cracks or contain any loose parts after being subjected to vibration for two hours in each of three mutually perpendicular axes at a frequency varying from 5 to 500 cps. The total amplitude shall be 0.01 inch or an applied vibratory acceleration of ± 5g, whichever is the limiting value. The frequency cycle may be continuous from 5 to 500 or may be in steps. However, the rate of change shall be such that a complete cycle will consume approximately 15 minutes. Following vibration, the device shall comply with the requirements as specified in 3.4.2.

3.4.6 Five foot drop.-The device, in a fixture shall comply with MIL-STD-331, Test No. 111 and the requirements as specified in 3.4.2.

3.4.7 Electrical properties (Arm Safe Device)

3.4.7.1 Insulation resistance (with all switches open).-The insulation resistance between each terminal to every other terminal and ground shall be 100 megohms minimum, when measured at 500 ± 50 volts, dc, with current limited to five milliamperes max.

3.4.7.2 Insulation resistance (after function and with all switches closed).-The insulation resistance between the following terminals, 1 & 2, 1 & 3, 1 & 4, 1 & ground, 2 & 3, 2 & 4, 2 & ground, 3 & 4, 3 & ground, and 4 & ground shall be 100 megohms minimum when measured at 500 ± 50 volts, dc, with current limited to five milliamperes max.

3.4.7.3 Circuit resistance (After function).-The circuit resistance of each of the four circuits shall not exceed 0.3 ohms maximum under the following conditions: open circuit voltage 5.. ± 0.5 volts, dc; current 100 ± 2 milliamperes, (circuits 1-8, 2-7, 3-5 and 4-6).

3.4.8 Weight.-The device shall not weigh more than 1.50 pounds.

3.4.9 Jolt.-The device shall comply with MIL-STD-331, Test No. 101.

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3.4.10 Jumble.-The device shall comply with MIL-STD-331, Test No. 102. The unit shall be jumbled in a fixture to protect the setting shaft.

3.4.11 Ten foot drop.-The device, in a fixture, with setback safety leaves released shall comply with MIL-STD-331, Test No. 103.

3.5 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications, and the following:

3.5.1 Metal defects.-All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.5.1.1 Firing arm assembly.-The firing arm assembly shall contain no fracture of the bend.

3.5.2 Plastic Defects.

3.5.2.1 Covers and rotors.-All components shall be free from fractures, crazing, burrs, flaking of gold plate of inserts, foreign inclusions and bubbles.

3.5.2.2 Cables.-All cables shall be free from blisters, creases, splits, or dents. The conductor path shall not be bare. The cables shall have a minimum of 0.010 plastic material along the outer edges.

3.5.2.3 Solvents.-No solvents containing aromatic hydrocarbons, chlorinated hydrocarbons, and alkaline substances shall be used in conjunction with any plastic parts.

3.5.3 Soldering.-The soldering shall be uniform and electrically conductive. The joints shall have a smooth metallic appearance and shall show no evidence of excessive solder, mounds (build-up above cover), presence of flux, foreign matter or having been overheated (slight discoloration is permissible) and insulation adjacent to the soldering area shall not be damaged. The soldering procedure and certification of soldering operators shall be in accordance with MIL-S-45743. A copy of the certification shall be furnished the processing activity.

3.5.4 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.5.5 Burr.-No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

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3.5.6 Surface finish.-The requirements for surface finish are as detailed on the applicable drawing.

3.5.7 Protective finish.-Protective finish shall be in accordance with the applicable specifications and drawings, and shall apply to all surfaces of the component regardless of configuration.

3.5.8 Foreign matter.-No part of assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter. The methods and agents used in processing and cleaning shall not adversely affect the finish or contaminate the parts.

3.6 Serialization.-A serial number shall be assigned and affixed by the contractor to each assembly or part when required by the applicable drawing. The serial number shall be in addition to the lot numbers, but in no event shall there be any possibility of duplication nor any indication of order or size of production quantity in either lot designation, serial number or any other number.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance Terms and Definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.2 First Article Sample

4.2.1 Initial production sample.-The first units of production shall be submitted as an initial production sample. The sample shall consist of the following:

180 (aggregate) component parts as delineated in 4.2.2
175 (aggregate) subassemblies as delineated in 4.2.3
38 Arm safe devices, less setting mechanism assy, as delineated in 4.2.5 and 4.2.6

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

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4.2.1.1 Examination and test. -After inspection by the contractor and provisional acceptance at source by the Government, one hundred and eighty (180) component parts, 175 subassemblies and 38 Arm Safe Devices ahall be inspected, as provided herein, at a Government laboratory or such other facility as specified in the contract.

4.2.2 Components

4.2.2.1 Twenty of each of the (8) components shall be tested as in 4.3.3.5 (Hardness), and 4.3.3.6 (Bend).

4.2.2.2 Twenty of each spring (1) shall be tested as in 4.3.3.3 (Load).

4.2.3 Subassemblies

4.2.3.1 Twenty-five of each of the (7) subassemblies shall be tested as in 4.3.3.4 (Static Load).

4.2.4 Acceptance criteria. -Failure of any component or subassembly to comply with the requirements of the applicable drawing shall be cause for rejection of the initial production sample.

4.2.5 Arm Safe Devices. -Thirty-eight arm safe devices, less setting mechanism assembly shall be subjected to the MIL-Standard tests and test conditions as follows:

Test	Quantity	Requirement	Method
Jolt	4	3.4.9	4.4.9.10
Jumble (in a fixture)	4	3.4.10	4.4.9.11
Ten foot drop (in a fixture)	4	3.4.11	4.4.9.12

4.2.5.1 Acceptance criteria. -Shall be as specified in the MIL-STD and any occurrence of the following in any arm safe device shall be cause for rejection of the initial production sample:

a. Failure of any of the circuits to remain open (continuity test).

b. Broken, deformed or displaced parts which render the device unsafe.

c. Escapement lever upright broken or fractured.

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4.2.6 Test Condition	Quantity	Requirement	Method
Low acceleration (ambient)	2	3.4.2	4.4.9.4
Low acceleration (+ 125° F)	4	3.4.2	4.4.9.4
Low acceleration (- 40° F)	4	3.4.2	4.4.9.4
High acceleration (ambient)	4	3.4.3	4.4.9.5
Thermal Shock	4	3.4.4	4.4.2
Vibration (in a fixture)	4	3.4.5	4.4.9.8
Five foot drop (in fixture)	4	3.4.6	4.4.9.9

4.2.6.1 Acceptance criteria. -The occurrence of any of the following shall be cause for rejection of the initial production sample:

a. A premature function (see 6.4)

b. Failure of more than 3 elements of the initial production sample.

c. Failure of more than 1 element in any device (see 6.2)

d. Failure of the device to comply with the timing accuracy.

4.3 Inspection Provisions

4.3.1 Submission of product. -The product shall be presented in accordance with MIL-STD-105.

4.3.2 Examination. -One hundred percent examination shall be performed for all critical defects. Examination for major defects shall be performed on a single defect basis in accordance with MIL-STD-105. For the classification of defects contained herein, the AQL given for each shall be used. All non-conforming materials shall be rejected. Four power (4X) magnification may be used in performing visual examinations for such defects as burr and foreign matter. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis of comparison for surface roughness determination.

4.3.2.1 Cable, Lamination No. 1, 2, 3, 4 Dwg (s) F10522724, 725, 726, 727 and F8596723 covering a detail of Dwg. E8596720 (defects listed are not specified on Dwg. F8596723).

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR AQL= 1.0 %	
101. Diameter of paths around contacts, improper	Gage
102. Path width, below min.	Gage
103. Path spacing, below min.	Gage

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4.3.2.1 (Cont'd)

Defect	Method of Inspection
MAJOR	
104. Distance from edge to path, below min.	Gage
105. Spacing between center of any contact and any path not connected to it, below min.	Gage
106. Location of holes, incorrect	Gage
107. Cable cut, nicked, gouged, bubbles or warping	Visual
108. Foreign matter	Visual

4.3.2.2 Cable, Laminated No. 5 Dwg F10522728 covering a detail of Dwg. E8596720.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR AQL=1.0%	
101. Location of holes, incorrect	Gage
102. Thickness, incorrect	Gage
103. Cable, cut, nicked or gouged	Visual
104. Foreign matter	Visual

4.3.2.3 Rotor Dwg.(s) D8597154 or C10522732 covering a detail of Dwg. E8596720.

Defect	Method of Inspection
CRITICAL	
1. Diameter of No. 1 hole (D8597154)	Gage
2. Location of No. 1 hole (D8597154)	Gage
3. Location of No. 1 radius (C10522732)	Gage
4. Diameter of rotor, min. (C10522732 or D8597154)	Gage
MAJOR AQL=0.25%	
101. Diameter of rotor, max. (both dwgs)	Gage
102. Angular width, improper (both dwgs)	Gage
103. Plastic defective (crack or burr)	Visual
104. Surface finish, improper	Visual

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4.3.2.4 Standard Cable Drawing D8597155, Laminated Cable Assembly D10522718 covering a detail of Drawing E8596720.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR AQL=1.0%	
101. Laminated in interconnecting area	Visual
102. Foreign matter	Visual

4.3.2.5 Switch Assembly Drawing F8596723 or Switch and Cable Assembly Drawing D10522714 covering a detail of Drawing E8596720 (prior to assembly of insulator assembly(s) and screws).

CRITICAL	
1. Protrusion of rivet(s) heads (manually, rotor hangs up on rivet or does not rotate freely) - (if applicable)	Manual/Visual
2. Torsion or actuator spring, missing (as applicable)	Visual
MAJOR AQL=0.25%	
101. Soldering, inadequate, missing, excessive or charred (cable on cover)	Visual
102. Foreign matter or presence of flux	Visual

4.3.2.6 Switch Assembly Drawing F8596723 or Switch and Cable Assembly Drawing D10522714 covering a detail of Drawing E8596720 (after assembly of insulator assemblies).

CRITICAL	
None defined	
MAJOR AQL=1.0%	
101. Gap excessive between covers	Visual
102. Contacts on terminal block damaged	Visual
103. Bubbles or warping (cables)	Visual
104. Foreign matter	Visual

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4.3.2.7 Movement Assembly Drawing F8596718 covering a detail of Dwg. E8596720.

Defect	Method of Inspection
CRITICAL	
1. Fracture of upright bend of firing arm	Visual
MAJOR AQL=0.40%	
101. Escapement lever upright(s) bent or broken	Visual
102. Firing arm spring, not hooked	Visual
103. Escapement lever not locked	Visual
104. Setback leaves, not in safe position	Visual
105. Any part missing	Visual
106. Foreign matter	Visual

4.3.2.8 Housing A-S Device Drawing F8596773 covering a detail of Dwg. E8596720.

CRITICAL	
None defined	
MAJOR AQL=1.0%	
101. Outside diameter, improper	Gage
102. Inside diameter, improper	Gage
103. Diameter of flange, improper	Gage
104. Total length, improper	Gage
105. Location of holes in flange, improper	Gage
106. Metal defective	Visual
107. Protective finish inadequate or defective	Visual

4.3.2.9 Arm Safe Device, less setting mechanism Drawing E8596720 (prior to assembly in housing).

CRITICAL	
1. Setting pinion torque above max.	Test
2. Assembly not set on zero	Visual
3. Any part missing	Visual
4. Improper assembly	Visual
MAJOR AQL=1.0%	
101. Locking screw not engaged properly	Visual

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4.3.2.10 Arm Safe Device, less setting mechanism Drawing E8596720.

Defect	Method of Inspection
CRITICAL	
1. Any part missing	Visual
2. Improper assembly	Visual
MAJOR AQL=1.0%	
101. Weight, incorrect	Balance
102. Terminal pin, missing, bent, loose or otherwise damaged	Manual and Visual
103. Case damaged	Visual
104. Marking missing, incorrect or illegible	Visual
105. Warning tag missing	Visual
106. Foreign matter	Visual

4.3.2.11 Packing and Marking (Level C).- The number of specimens and acceptance criteria for packing and marking shall be in accordance with MIL-P-10025.

4.3.2.12 Monitoring - Periodic unscheduled monitoring shall be conducted for the process operations on the following to determine compliance with the requirements of the applicable drawings.

Soldering techniques
Solvents - cleaning plastic parts
Laminating conditions - Drawing D10522718

Drawing F8596718

Winding of mainspring, torque
Setting gear, torque
Timing disc, torque
Regulation
Stake retainer plate No. 3
Application of sealant
Pinion(s) escapement No. 1, 2, 3, 4 (lubrication)
Escapement arbor
Setback safety arbor
Firing arm shaft

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Protective finish.- The number of specimens and acceptance criteria for protective finish tests shall be in accordance with the applicable standard or specification listed on the drawing.

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4.3.3.2 Plastic materials.

4.3.3.2.1 Thermal shock.-Two each of the following: rotors, top cover, bottom cover, shall be selected at random from each 100 or less of each part in the lot and subjected to thermal shock. Failure of any part to comply with the requirements of 3.1.1.1 shall be cause for rejection of the lot of plastic material.

4.3.3.2.2 Ceritification.-Each batch of plastic material shall be certified as meeting the requirements of the applicable plastic specification and MIL-STD. A copy of the certification shall be furnished the Government.

4.3.3.3 Load.-Twenty springs from each lot shall be subjected to the load test. Failure of any component to comply with the requirements of Dwg. B10522729 shall be cause for rejection of the lot. The test shall be performed in accordance with 4.4.4.

4.3.3.4 Static load.-Twenty-five samples from each lot of the following sub-assemblies shall be subjected to the static load test. Failure of any sub-assembly to comply with the requirements specified on the applicable drawing shall be cause for rejection of the lot of sub-assemblies, as applicable (see 4.4.6).

Assembly	Drawing
No. 1 gear and pinion assembly	C8596758
No. 2 gear and pinion assembly	C8596759
No. 3 gear and pinion assembly	C8596760
No. 4 gear and pinion assembly	C8596761
Escapement gear and pinion assembly	C8596762
Safety arm and stud assembly	C8596763
Leaf and arbor, setback safety assembly	C8596770

4.3.3.5 Hardness.-Twenty samples from each lot of the following components shall be tested for hardness. Failure of any component to comply with the requirements specified on the applicable drawing shall be cause for rejection of the lot. The test shall be performed in accordance with paragraph 4.4.5.

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Components	Drawing
Washer	B8596794
Clip, safety arm	C8596789
Washer	C8596795
Washer, spring, timing disc safety	C10522938
Washer, spring	C10522945
Arm, firing No. 2(See 4.3.3.6)	C10522871
Guide, adjusting	B10522959
Lever, escapement	D8596730

4.3.3.6 Firing arm bend test. -The same 20 firing arms that comply with the hardness reqirements shall be subjected to the bend test. Evidence of fracture after bending (upright to bend 20 degrees, minimum) shall be cause for rejection of the lot (See 4.4.7).

4.3.3.7 Movement Assembly

4.3.3.7.1 Regulation. -Each movement shall be subjected to 2 consecutive runs for 50 seconds. Any movement failing to comply with the requirements of 3.3.1 shall be rejected. The test shall be performed in accordance with 4.4.8.1.

4.3.3.7.2 Non-Arming. -Each movement shall be subjected to a nonarming test. Any movement failing to comply with the requirements of 3.3.2 shall be rejected. The test shall be performed in accordance with 4.4.8.4.

4.3.3.7.3. -Each movement shall be subjected to an arming test. Any movement failing to comply with the requirements of 3.3.3 shall be rejected. The test shall be performed in accordance with 4.4.8.3.

4.3.3.7.4 Function, 1150 to 1300g. -Each movement shall be subjected to 2 consecutive function and running tests. Any movement failing to comply with the requirements of 3.3.4 shall be rejected. The test shall be performed in accordance with 4.4.8.2.

4.3.3.8 Switch Assembly

4.3.3.8.1 Insulation resistance, with all switches closed. -Each switch assembly shall be subjected to an insulation test. Any switch failing to comply with the requirements of 3.3.1 shall be rejected. Test shall be performed in accordance with 4.4.3.1.

4.3.3.8.2 Circuit resistance. -The three switches and all combinations of 2 of the 3 switches shall be subjected to a circuit resistance test. Any switch failing to comply with the requirement of 3.2.2 shall be rejected. The test shall be performed in accordance with 4.4.3.2.

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4.3.3.9 Arm Safe Device, Less Setting Mechanism

4.3.3.9.1 Function (prior to assembly in housing). -Each device shall be checked to determine the initial setback leaf is free to rotate when the device is set at 1 second, min. Any device with the initial setback leaf restricted shall be rejected. The test shall be performed in accordance with 4.4.9.1.

4.3.3.9.2 Safety (prior to assembly in housing). -Each device shall be checked to determine that the initial setback leaf is not free to rotate when the device is set at 1/2 second and at zero (0) second settings. Any device with an unrestricted initial setback leaf shall be rejected. The test shall be performed in accordance with 4.4.9.1.

4.3.3.9.3 Insulation resistance, with all switches open. -Each device shall be subjected to an insulation resistance test. Any device failing to comply with the requirement of 3.4.7.1 shall be rejected. The test shall be performed in accordance with 4.4.9.2.1.

4.3.3.10 Production samples. -Twelve devices shall be selected from each lot and subjected to the following tests.

Condition	Quantity	Requirement	Method
Low acceleration (ambient)	2	3.4.2	4.4.9.4
Low acceleration (+125° F)	2	3.4.2	4.4.9.4
Low acceleration (-40° F)	4	3.4.2	4.4.9.4
Vibration (in a fixture)	2	3.4.5	4.4.9.8
Five foot drop (in a fixture)	2	3.4.6	4.4.9.9

Acceptance criteria: A premature function or failure of more than one element to comply with applicable requirements shall be cause for rejection of the lot (See 6.2 and 6.4).

4.3.4 Inspection equipment. -The examination and tests shall be made using equipment listed in IEL 8596720. Equipment design shall be in accordance with the code designation on the applicable IEL.

4.4 Test Methods and Procedures

4.4.1 Protective finish. - Test procedures for protective finish shall be in accordance with the specification or standard referenced on the component drawing.

4.4.2 Thermal shock (Destructive). -The thermal shock test shall be in accordance with MIL-STD-331, test No. 113. A light table with four power (4X) magnification shall be used in performing visual examination of the rotors and covers. Subsequent to the thermal shock test, the device shall be tested as specified in 4.4.9.3.

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4.4.3 Switch assembly (Non-destructive).

4.4.3.1 Insulation resistance (with all switches closed). -The insulation resistance between the following terminals: 1 & 2, 1 & 3, 1 & 4, 2 & 3, 2 & 4, and 3 & 4 shall be 100 megohms, min., when measured at 500 ± 50 volts, dc, with current limited to five milliamperes, max. The opening and closing of the switches are done by means of probes.

4.4.3.2 Circuit Resistance. -Measured at 100 ± 5 milliamperes. -Prior to testing, the open circuit voltage of the output of the test equipment shall be 5.5 ± 0.5 volts, dc. The resistance of each closed circuit shall not exceed 0.3 ohms, max.

Switches	Circuits	Switches	Circuits
1, 2 and 3	1-8 2-7 3-5 4-6	1 and 3	1-8 2-7 3-5 4-6
1 and 2	1-8 2-7 3-5 4-6	2 and 3	1-8 2-7 3-5 4-6

4.4.4 Load (Nondestructive). -The springs shall be compressed to the height and the load applied as specified on the applicable drawing.

4.4.5 Hardness (Destructive). -Samples shall be tested in an area which is solidly supported so as to obtain valid readings. The median value of 3 hardness readings taken on each sample unit shall be used as the individual value.

4.4.6 Static load. -Escapement lever assemblies shall be tested prior to insertion of the escapement lever spring. The test shall be performed using the values as specified on the applicable drawing. All samples tested shall be scrapped.

4.4.7 Firing arm bend (Destructive). -All firing arms that have been subjected to a bend test (upright to bend 20 degrees, min.) shall be scrapped.

4.4.8 Movement Assembly (Nondestructive).

4.4.8.1 Static regulation. -The movement shall be regulated using the operational procedures furnished with the equipment. Timekeeping and performance shall be checked for a period of not less than 50 seconds. No adjustment shall be made on the escapement during the 50 second check runs. Inspection shall be performed on a monitoring basis.

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4.4.8.2 Function (movement assembly with setback leaf released. -The movement assembly shall be mounted on a fixture which is secured to the load column of an 18 inch Hyge Tester. The test shall be performed twice at a linear acceleration of 1150 to 1300g to determine function of setback mechanism, running of movements and release of firing arm. After completion of each test, the movements shall be rewound and the setback mechanism reset for a nonarming test.

4.4.8.3 Arming (movement with setback safety leaf released). -The movement assembly (several) shall be mounted on a fixture, secured to the table of a centrifuge. The centrifuge shall be rotated to a speed required to obtain 860g max., to determine release of the escapement lever lock. After the arming test the setback mechanism shall be reset and the movement rewound.

4.4.8.4 Nonarming (movement, with setback safety leaf released). -The escapement lever lock shall not release at an acceleration of 600g, minimum. The non-arming test shall be performed after the function and arming tests.

4.4.9 Arm Safe Device, Less Setting Mechanism

4.4.9.1 Function and safety (prior to assembly in housing). -The device shall be assembled into a simulated setting fixture, set at one second and checked to be sure that the safety setback leaf is not preventing the initial setback leaf from rotating freely. Check the position of the timing disc slot with respect to the safety arm to insure function at one second setting. The device shall then be set and checked at ½ second and zero (0) setting to be sure that the safety leaf is preventing the initial setback leaf from rotating.

4.4.9.2 Electrical Properties (See 3.4.7) Arm Safe Device.

4.4.9.2.1 Insulation resistance (prior to function of device and with all switches open) 100 megohms, min. when measured at 500 ± 50 volts, dc, with current limited to five milliamperes, max.

Connection	Connection	Connection
Ground to 1	1 to 2	2 to 3
Ground to 2	1 to 3	2 to 4
Ground to 3	1 to 4	2 to 5
Ground to 4	1 to 5	2 to 6
Ground to 5	1 to 6	2 to 7
Ground to 6	1 to 7	2 to 8
Ground to 7	1 to 8	
Ground to 8		

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Connection	Connection	Connection	Connection
3 to 4	4 to 5	5 to 6	6 to 7
3 to 5	4 to 6	5 to 7	6 to 8
3 to 6	4 to 7	5 to 8	7 to 8
3 to 7	4 to 8		
3 to 8			

4.4.9.2.2 Insulation and circuit resistance. -After function and all switches closed (See 4.4.9.3, 3.4.7.2 and 3.4.7.3).

4.4.9.3 Arm safe device, less setting mechanism (sequence and additional procedures). -The device less safety screw, shall be assembled into a simulated setting fixture and set at 45 seconds. After disassembly from the setting fixture, the following shall be performed. The component parts of the arm safe device, which have been disassembled and inspected following tests, shall be scrapped.

a. Activate with 1150 to 1300g acceleration (18 inch Hyge) to release the escapement lever lock.

b. Record functioning time to determine timing accuracy in accordance with requirement 3.4.1.2 (See IEL).

c. Measure resistance of each circuit with the switches in the operated condition (closed) (See 3.4.7.2 and 3.4.7.3).

d. Disassemble device, examine and record the following:

(1) Functioning of each movement (ran fully, partially ran or failed to start)
(2) Functioning of each setback mechanism
(3) Function of firing arms
(4) Position of each switch (open or closed)

4.4.9.4 Low Acceleration (destructive). -The device shall be mounted on a fixture which is secured to the load column of an 18 inch Hyge Tester and activated with 1150 to 1300g linear acceleration. The operating procedure shall be outlined by the manufacturer. The device shall be temperature conditioned and tested as specified in 4.2.6 and 4.3.3.10. (Test sequence and procedure - see 4.4.9.3).

4.4.9.5 High acceleration (destructive). -The device shall be activated with 2500 to 2750g linear acceleration (18 inch Hyge) and tested as specified in 4.2.6 and 4.4.9.3 with the activation change.

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4.4.9.6 Low temperature(-40° F destructive). -The test chamber shall be at ambient temperature at the start of the test. The temperature shall be reduced at a uniform rate with the device in the chamber. When the low temperature of minus 40 degrees (F) is reached after which the device shall be subjected to minus 40 degrees (F) for 1 hour, minimum. The device shall be removed from the conditioning chamber and within a period of 5 minutes operated in accordance with the procedures in 4.4.9.3.

4.4.9.7 High temperature (+ 125° F-destructive). -The test chamber shall be at ambient temperature at the start of the test. The temperature shall be increased at a uniform rate with the device in the chamber. When the high temperature of plus 125 degrees (F) is reached after which the device shall be subjected to plus 125 degrees (F) for 1 hour, minimum. The device shall be removed from the conditioning chamber and within a period of 5 minutes operated in accordance with the procedures in 4.4.9.3.

4.4.9.8 Vibration (in a fixture-destructive). -The vibration test shall be performed in accordance with 3.4.5 using equipment in accordance with 4.3.4. After vibration, the device shall be operated in accordance with the procedures in 4.4.9.4.

4.4.9.9 Five foot drop. -(In a fixture-destructive). -The test shall be performed in accordance with MIL-STD-331 test No. 111 and dropped basedown. After the drop the device shall be operated in accordance with the procedures in 4.4.9.4.

4.4.9.10 Jolt (in a fixture-destructive). -The test shall be performed in accordance with MIL-STD-331, test No. 101.

4.4.9.11 Jumble (in a container-destructive).-The test shall be performed in accordance with MIL-STD-331, test No. 102. A container shall be used to protect the setting shaft.

4.4.9.12 Ten foot drop (in a fixture-destructive). -The test shall be performed in accordance with MIL-STD-331, test No. 103, except that the number of units as specified in 4.2.5 shall be used in lieu of five. All units shall be dropped base down (setting shaft down) and with the setback safety leaves released.

5. PREPARATION FOR DELIVERY

5.1 Level A - Not applicable.

5.2 Level B - Not applicable.

5.3 Level C - Packaging and Marking.-Unless otherwise specified packaging and marking shall be in accordance with MIL-STD-1169.

5.4 Data Cards. -Data card information shall be as specified in MIL-STD-1167.

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6. NOTES

6.1 Components manufactured in strict accordance with applicable drawings represent one set of conditions which will result in satisfactory performance. Dimensions which are specifically required by classification of defects as listed in Section 4 or those which affect external profile of assembly are mandatory. Pertaining to other dimensions, it is recognized that current mass production methods may occasionally produce parts which are not in full compliance with all dimensional requirements, but which may provide adequate performance. Therefore, the contractor, at his own risk, may utilize these parts provided there is no adverse effect on device function-ability. This relaxation of requirements applies to "after the fact" deviations only, to dimensions only, on applicable drawings, and does not constitute authority to establish new dimensions or to relax efforts to achieve the specified limits. The procedure for accomplishing dimensional changes is available under the established engineering change system. Nothing above is to be construed as relieving the contractor of any responsibility in connection with performance of the fuze.

6.1.1 The contractor shall maintain records showing deviations from dimensional requirements, the extent of such deviations, the amount of material involved, and the date the material is introduced into the assembly line. Copies of these records shall be forwarded to the agency assigned custody of the technical data.

6.2 An element is considered to be a movement assembly, a switch assembly or any other part which contributes to the desired end result of each switching function being satisfactorily accomplished.

6.3 Ambient as defined herein is room temperature $70^{\circ} \pm 10^{\circ}$ Fahrenheit.

6.4 Premature function - Premature function is when the device:

a. Set for 1 second, functions at less than 0.5 seconds of the time set
b. Set for 2 to 20 seconds, functions at less than 75% of the time set
c. Set for 21 to 45 seconds, functions at less than 5 seconds of the time set

If a premature occurs, the contractor and the Government inspector will conduct an investigation to determine the cause of the premature and take necessary action to prevent any repetition.

6.5 Ordering data. -Procurement documents should specify the title, number, and date of this specification.

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6.6 Supersession data. -This specification includes the requirements of Purchase Description FA-PD-MI-2395, Rev. 0 dated 29 July 1960.

CUSTODIAN:

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PREPARING ACTIVITY:

Army-MU

Project No. 1390 - A721

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